

8 The types of lipid consumed in the diet can affect the health of individuals.

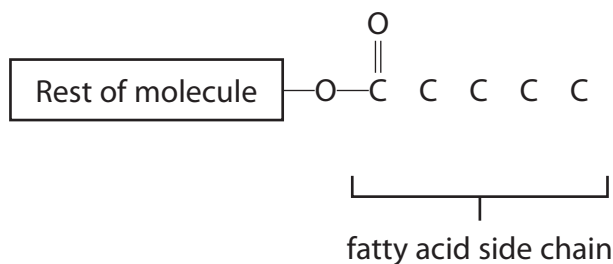
(a) Tropical sprue is a disorder that limits the absorption of nutrients into the blood.

Consumption of unsaturated lipids has been proposed as a cause of tropical sprue.

The diagram shows part of an unsaturated lipid.

Complete the diagram to show one possible structure of a fatty acid side chain of an unsaturated lipid.

(2)



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(b) Consumption of high levels of cholesterol is associated with atherosclerosis.

In the development of atherosclerosis, lesions form in the coronary arteries and these develop into atheromas (plaques).

In an investigation, one group of primates was fed a diet supplemented with cholesterol and saturated fatty acids.

A second group was fed a diet supplemented with cholesterol and unsaturated fatty acids.

The primates were fed these diets from a young age and then monitored over a period of 60 months.

Primates are a higher order of mammals that include monkeys.



The results of the investigation are shown in the graphs.

Key

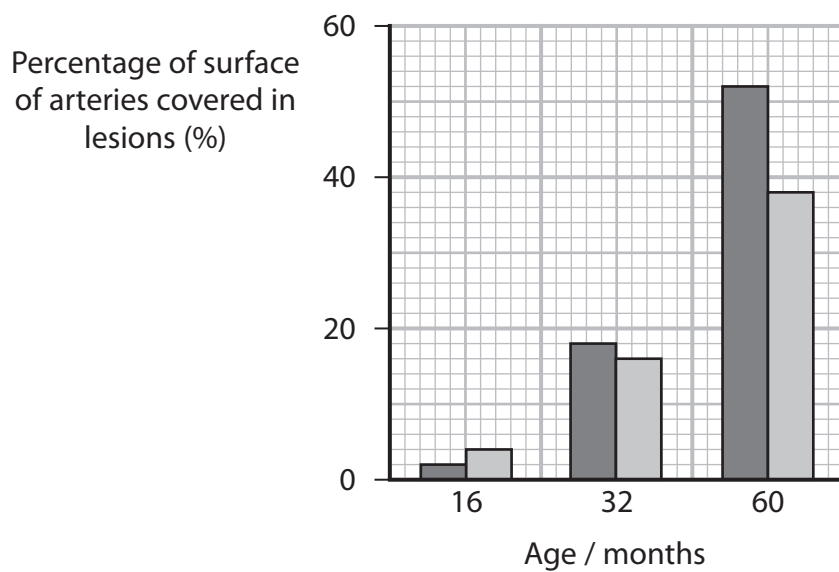


saturated lipids

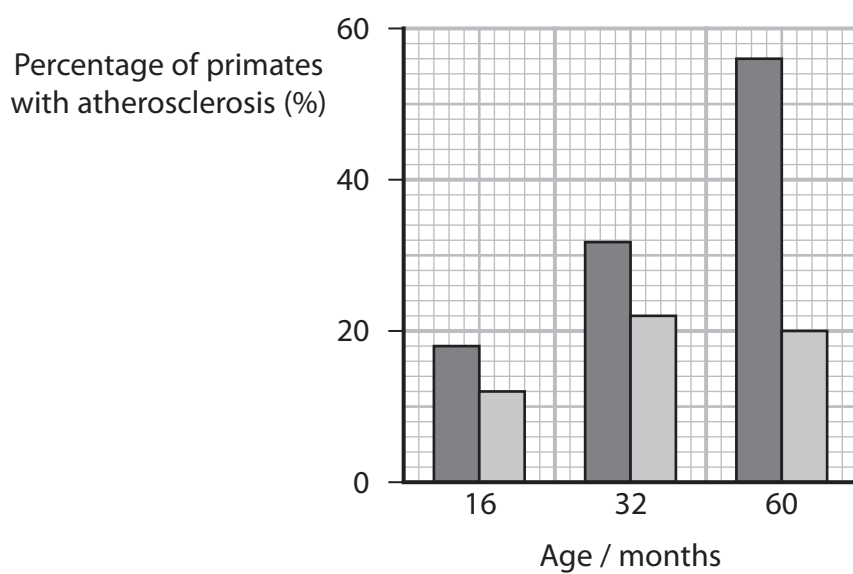


unsaturated lipids

Graph 1



Graph 2



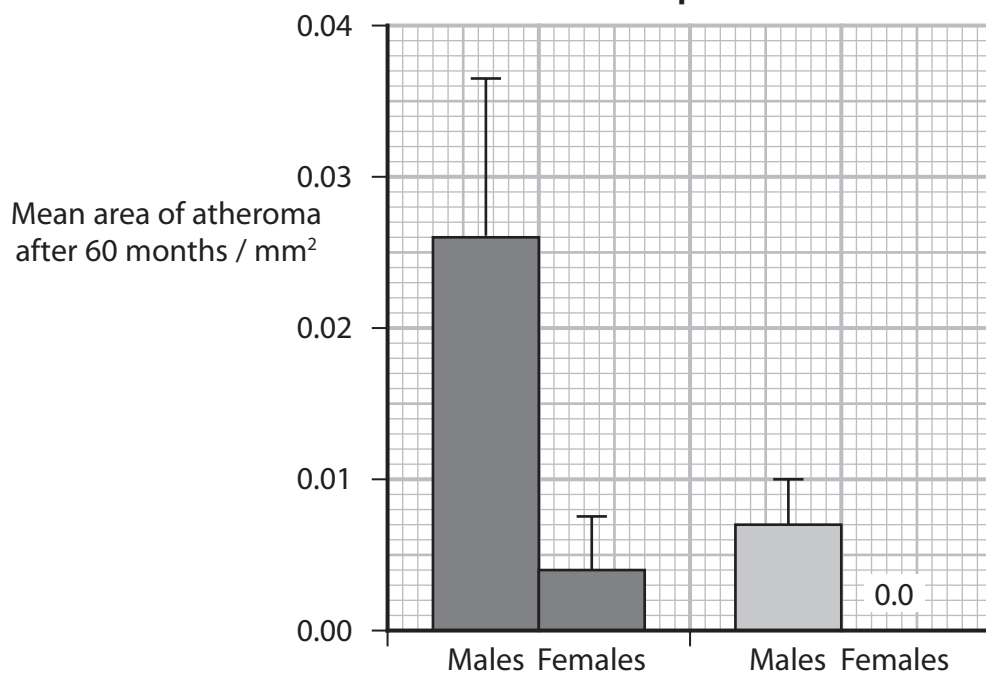
Key



saturated lipids



unsaturated lipids

Graph 3

- (i) Calculate the mean rate of increase in the surface of arteries covered in lesions from 16 to 60 months in the primates given saturated lipids (graph 1).

(1)

Answer percentage per month

- (ii) Calculate the ratio of primates with atherosclerosis given saturated lipids to those given unsaturated lipids, at 32 months (graph 2).

(1)

Answer



- [illegible]



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- (v) Discuss the ethical issues relating to the use of primates and other animals in investigations such as this one.

(3)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS

